QUBO Formulation: Max-Cut

Max-Cut

Problem Statement. Given an undirected graph $G = (V, E)$ with $|V| = N$ nodes and a set of weighted edges E, where each edge $(i, j) \in E$ carries a non-negative weight w_{ij}, the Max-Cut problem seeks to partition the vertex set V into two disjoint subsets S_0 and S_1 such that the total weight of edges connecting a node in S_0 to a node in S_1 is maximized.

Decision Variables. For each node $i \in V$, we introduce a binary decision variable $x_i \in \{0, 1\}$. The variable x_i encodes the group assignment of node i, where $x_i = 0$ indicates membership in subset S_0 and $x_i = 1$ indicates membership in subset S_1. The complete assignment is represented by the vector $\mathbf{x} = (x_0, x_1, \ldots, x_{N-1})^\top \in \{0, 1\}^N$.

QUBO Derivation. Step 1 — Objective Formulation. The weight of the cut induced by an assignment $\mathbf{x}$ is the sum of weights of edges crossing between the two partitions. An edge $(i, j) \in E$ crosses the cut if and only if $x_i \neq x_j$. This condition is captured by the indicator function $x_i + x_j - 2x_ix_j$, which evaluates to 1 when $x_i \neq x_j$ and 0 otherwise. The original maximization objective is therefore

$$\max_{\mathbf{x}\in\{0,1\}^N} \sum_{(i,j)\in E} w_{ij}(x_i + x_j - 2x_ix_j). \tag{1}$$

Since QUBO solvers minimize energy, we negate the objective to obtain the minimization problem:

$$\min_{\mathbf{x}\in\{0,1\}^N} H(\mathbf{x}) = \min_{\mathbf{x}\in\{0,1\}^N} \left(- \sum_{(i,j)\in E} w_{ij}(x_i + x_j - 2x_ix_j) \right). \tag{2}$$

Step 2 — Constraints. The Max-Cut problem imposes no additional constraints on the binary variables; any assignment $\mathbf{x} \in \{0, 1\}^N$ corresponds to a valid partition. Thus, the feasible region is the entire Boolean hypercube.

Step 3 — Penalty Terms. As there are no constraints, no penalty terms are required. The Hamiltonian consists solely of the negated objective.

Step 4 — Combined Hamiltonian. Expanding the negated objective and grouping terms by variable powers yields:

$$H(\mathbf{x}) = -\sum_{(i,j)\in E} w_{ij}x_i - \sum_{(i,j)\in E} w_{ij}x_j + 2\sum_{(i,j)\in E} w_{ij}x_ix_j. \tag{3}$$

We collect the linear coefficients for each node $i \in V$. Each node i appears in the first sum for every edge incident to i, and in the second sum for every edge incident to i. Let $d_i = \sum_{j:(i,j)\in E} w_{ij}$ denote the weighted degree of node i. The linear part simplifies to $-\sum_{i\in V} d_i x_i$. The quadratic part remains $2\sum_{(i,j)\in E} w_{ij}x_ix_j$. Thus, the full QUBO Hamiltonian is

$$H(\mathbf{x}) = \sum_{i\in V}\left(-\sum_{j:(i,j)\in E} w_{ij}\right)x_i + \sum_{(i,j)\in E} 2w_{ij}x_ix_j. \tag{4}$$

Step 5 — Q Matrix and Offset. The QUBO Hamiltonian is generally expressed as $H(\mathbf{x}) = \mathbf{x}^\top Q\mathbf{x} + c$. Comparing with the expanded form above, we identify the diagonal and off-diagonal entries of the symmetric matrix Q and the constant offset c:

$$Q_{i,i} = -\sum_{j:(i,j)\in E} w_{ij}, \qquad \forall i \in V, \tag{5}$$

$$Q_{i,j} = 2w_{ij}, \qquad \forall (i,j) \in E,\ i \neq j, \tag{6}$$

$$Q_{i,j} = 0, \qquad \forall (i,j) \notin E,\ i \neq j, \tag{7}$$

$$c = 0. \tag{8}$$

Penalty Weight Justification. In constrained QUBO reductions, a penalty weight λ is typically chosen to exceed the maximum possible objective gain from violating a constraint, ensuring that any energy reduction from violating a constraint is outweighed by the penalty term. For the Max-Cut problem, there are no constraints; every binary vector $\mathbf{x} \in \{0,1\}^N$ corresponds to a valid graph partition. Consequently, the feasible region is unconstrained, no penalty terms are required, and the Hamiltonian directly encodes the optimization landscape over the full hypercube without the need for a penalty weight.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0,1\}^N$ recovers the optimal Max-Cut partition because the mapping between binary assignments and graph cuts is bijective and unconstrained. Every $\mathbf{x}$ yields a valid cut, and the Hamiltonian is constructed such that $H(\mathbf{x}) = -\text{CutWeight}(\mathbf{x})$. Therefore, the global minimum of H corresponds exactly to the assignment that maximizes the cut weight. Since the feasible set is the entire Boolean hypercube, no infeasible solutions exist to penalize, and the ground state of H is guaranteed to be an optimal solution to the original Max-Cut problem.

Illustrative Example. Consider a concrete instance with $N = 3$ nodes and a complete graph K_3 where $E = \{(0,1),(0,2),(1,2)\}$. Assign concrete edge weights $w_{01} = 1$, $w_{02} = 2$,

and $w_{12} = 3$. Substituting these values into the general formulas yields the following concrete objective and Hamiltonian:

$$H(\mathbf{x}) = (-1-2)x_0 + (-1-3)x_1 + (-2-3)x_2 + 2(1)x_0x_1 + 2(2)x_0x_2 + 2(3)x_1x_2 \tag{9}$$
$$= -3x_0 - 4x_1 - 5x_2 + 2x_0x_1 + 4x_0x_2 + 6x_1x_2. \tag{10}$$

The corresponding symmetric Q matrix entries and offset are:

$$Q_{0,0} = -3, \qquad Q_{1,1} = -4, \qquad Q_{2,2} = -5, \tag{11}$$
$$Q_{0,1} = 2, \qquad Q_{0,2} = 4, \qquad Q_{1,2} = 3, \tag{12}$$
$$c = 0. \tag{13}$$

Enumerating all $2^3 = 8$ binary assignments reveals the ground state at $\mathbf{x}^* = (0, 0, 1)^\top$ with energy $H(\mathbf{x}^*) = -5$. This corresponds to partitioning node 2 into group 1 and nodes $0, 1$ into group 0, yielding a maximum cut weight of 5 (edges $(0, 2)$ and $(1, 2)$).